Calculus I — Limits & Derivatives

domain blockers

Ban exactly what breaks: **denominators** $\neq 0$, **even roots** need $(\cdot) \geq 0$, **logs** need $(\cdot)>0$. Every domain question from here through multivariable is these three checks.

slope

$m=\frac{y_2-y_1}{x_2-x_1}$ $\quad y-y_1=m(x-x_1)$ $\quad y=mx+b$

avg / instantaneous rate of change

$AV_{[a,b]}=\frac{f(b)-f(a)}{b-a}$ $\quad \lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$

limits

$\lim_{x\to a} f(x)=L$ $\quad \lim_{x\to a^-} f(x)=L$ $\quad \lim_{x\to a^+} f(x)=L$
Infinite limit: if $f(x) \to \pm\infty$ as $x \to a$, write $\lim_{x\to a} f(x)=\pm\infty$ (vertical asymptote at $x=a$) — no finite limit. The two-sided limit exists only if both one-sided limits agree.

Which technique? — try them in this order

1. **Substitute.** If you get a number, that is the limit. Always try this first.
2. Got $\frac{0}{0} \Rightarrow$ **algebraic techniques**: factor and cancel, multiply by the conjugate, or clear a complex fraction.
3. Still $\frac{0}{0}$ or $\frac{\infty}{\infty} \Rightarrow$ **L'Hôpital's rule**. Differentiate top and bottom separately.
4. Trapped between two bounds, or a $\sin\frac{1}{x}$ factor $\Rightarrow$ **squeeze theorem**.
5. $x \to \pm\infty$ on a rational function $\Rightarrow$ **compare leading exponents**.
6. A piecewise split, or $|x|$ at its zero $\Rightarrow$ **one-sided limits**. They must agree.

A number over zero is not $\frac{0}{0}$ — that is an infinite limit or a vertical asymptote, and no technique above applies.

Worked: one-sided limits on a piecewise function

$$f(x)=\begin{cases} x^2 & x<1 \\ 3x+1 & x \geq 1 \end{cases} \quad \text{at } x=1\text{:}$$

1. LEFT — USE THE $x<1$ PIECE
$\lim_{x\to 1^-} f=1^2=1$

↓

2. RIGHT — USE THE $x \geq 1$ PIECE
$\lim_{x\to 1^+} f=3(1)+1=4$

↓

3. COMPARE
$1 \neq 4 \Rightarrow \lim_{x\to 1} f(x)$ does not exist (jump)

limits from leading exponents

$$\lim_{x\to\infty}\frac{cx^a}{dx^b}=\begin{cases} +\infty & a>b,\ c/d>0 \\ -\infty & a>b,\ c/d<0 \\ 0 & a<b \\ \frac{c}{d} & a=b \end{cases}$$

One per case: $\lim_{x\to\infty}\frac{3x^2+1}{x^2-5}=3$ $(a=b)$; $\quad \lim_{x\to\infty}\frac{x+1}{x^2}=0$ $(a<b)$;
$\lim_{x\to\infty}\frac{x^3}{x^2}=+\infty$ $(a>b)$.

At $-\infty$ watch odd powers: $\lim_{x\to-\infty}\frac{x^3}{x^2}=\lim_{x\to-\infty}x=-\infty$ — re-derive the sign, don't copy the $+\infty$ case.

asymptotes — the finding procedure

- **Vertical**: zeros of the denominator that survive cancelling — a cancelled zero is a hole, not a VA.
- **Horizontal**: $y=\lim_{x\to\pm\infty} f$, via leading exponents (above).
- **Slant**: when $deg\,(\text{top})=deg\,(\text{bottom})+1$ — long-divide. The quotient line is the asymptote.

Worked: all asymptotes of $f=\frac{2x^2+3x-1}{x^2-4}$

1. VERTICAL
$x^2-4=0$, nothing cancels with the top: $x=\pm 2$

↓

2. HORIZONTAL
degrees equal $\Rightarrow$ ratio of leading coefficients: $y=2$

Worked: slant asymptote of $f=\frac{x^2+1}{x-1}$

1. DEGREE CHECK
$2=1+1 \Rightarrow$ slant exists (and no HA)

↓

2. LONG DIVIDE
$x^2 \div x=x$; $\quad x\cdot(x-1)=x^2-x$; subtract: remainder $x+1$. Then $x \div x=1$; $\quad 1\cdot(x-1)=x-1$; subtract: remainder 2:

$$\frac{x^2+1}{x-1}=x+1+\frac{2}{x-1}$$

(check: $(x-1)(x+1)+2=x^2-1+2=x^2+1$ ✓*)*

↓

3. DROP THE REMAINDER
$y=x+1$ $\quad$ *(the* $\frac{2}{x-1}$ *piece* $\to 0$ *at* $\pm\infty$*)*

Root trap at $-\infty$: $\sqrt{x^2}=|x|=-x$ for $x<0$:
$\frac{\sqrt{x^2+1}}{x}=\frac{|x|\sqrt{1+1/x^2}}{x}=-\sqrt{1+1/x^2}$, so $\lim_{x\to-\infty}\frac{\sqrt{x^2+1}}{x}=-1$, not $+1$.

fundamental limits

$\lim_{h\to 0}\frac{\sin(h)}{h}=1$ $\quad \lim_{h\to 0}\frac{\cos(h)-1}{h}=0$
$\lim_{h\to 0}\frac{e^h-1}{h}=1$ $\quad \lim_{h\to 0}\frac{\ln(1+h)}{h}=1$
$\lim_{h\to 0}(1+h)^{1/h}=e$
Rescaling: the formulas demand *matching* insides. $\lim_{x\to 0}\frac{\sin(3x)}{x}$: force the match — $\frac{\sin(3x)}{x}=3\cdot\frac{\sin(3x)}{3x} \to 3\cdot 1=3$.

limit laws

If $\lim_{x\to a} f$ and $\lim_{x\to a} g$ both exist:
- $\lim[f \pm g]=\lim f \pm \lim g$
- $\lim[c\cdot f]=c\cdot\lim f$
- $\lim[f\cdot g]=\lim f\cdot\lim g$
- $\lim\frac{f}{g}=\frac{\lim f}{\lim g}$, if $\lim g \neq 0$
- $\lim[f]^n=[\lim f]^n$

algebraic limit techniques — killing a $0/0$ by hand

Substitution gave $\frac{0}{0}$? That's indeterminate, not an answer — one of three algebra moves usually rescues it, no L'Hôpital required.

Worked: factor & cancel — $\lim_{x\to 3}\frac{x^2-9}{x-3}$

1. SUBSTITUTE
$\frac{9-9}{3-3}=\frac{0}{0}$ — indeterminate, keep going

↓

2. FACTOR & CANCEL
$$\frac{(x-3)(x+3)}{x-3}=x+3$$

↓

3. SUBSTITUTE AGAIN
$\lim_{x\to 3}(x+3)=6$

***Worked: conjugate* — $\lim_{x\to 0}\frac{\sqrt{x+4}-2}{x}$**

1. MULTIPLY BY THE CONJUGATE
$\cdot\frac{\sqrt{x+4}+2}{\sqrt{x+4}+2}$: the top becomes $(x+4)-4=x$

↓

2. CANCEL THE x
$$\frac{x}{x(\sqrt{x+4}+2)}=\frac{1}{\sqrt{x+4}+2}$$

↓

3. SUBSTITUTE
$$\frac{1}{\sqrt{4}+2}=\frac{1}{4}$$

***Worked: complex fraction* — $\lim_{h\to 0}\frac{\frac{1}{2+h}-\frac{1}{2}}{h}$**

1. COMBINE THE TOP
$$\frac{2-(2+h)}{2(2+h)}=\frac{-h}{2(2+h)}$$

↓

2. DIVIDE BY h
$$\frac{-1}{2(2+h)}$$

↓

3. SUBSTITUTE
$$\frac{-1}{2(2+0)}=-\frac{1}{4}$$ *(this is exactly $f'(2)$ for $f=1/x$ — the shape shows up in derivative-by-definition problems)*

Absolute value forces one-sided: $\lim_{x\to 0}\frac{|x|}{x}$ — left: $|x|=-x\Rightarrow -1$; right: $|x|=x\Rightarrow 1$; disagree ⇒ DNE. Any $|\cdot|$ at its zero point is a split-and-compare.

squeeze theorem

If $g(x)\le f(x)\le h(x)$ near a and $\lim_{x\to a}g=\lim_{x\to a}h=L$, then $\lim_{x\to a}f=L$.

***Worked:* $\lim_{x\to 0}x^2\sin(1/x)$**

1. BOUND THE WILD PART
$-1\le\sin(1/x)\le 1$ for all $x\neq 0$

↓

2. MULTIPLY BY x^2
$-x^2\le x^2\sin(1/x)\le x^2$ *($x^2\ge 0$ keeps the inequalities pointed the same way)*

↓

3. SQUEEZE
both fences $\to 0\Rightarrow\lim=0$ *(substitution and L'H are useless here — $\sin(1/x)$ has no limit)*

L'Hôpital's rule

Only when the form is indeterminate $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$ — check first. Then differentiate top and bottom *separately* (not the quotient rule):

$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}$

Once: $\lim_{x\to 0}\frac{\sin x}{x}$ — check $\frac{0}{0}$ ✓ — $=\lim\frac{\cos x}{1}=1$.

***Worked: applied twice* — $\lim_{x\to 0}\frac{e^x-1-x}{x^2}$**

1. CHECK THE FORM
$\frac{1-1-0}{0}=\frac{0}{0}$ ✓

↓

2. L'H ONCE — RE-CHECK
$\lim\frac{e^x-1}{2x}$: still $\frac{0}{0}$ — allowed to go again

↓

3. L'H AGAIN
$$\lim\frac{e^x}{2}=\frac{1}{2}$$

Non-indeterminate = hands off: $\lim_{x\to 0^+}\frac{\cos x}{x}=\frac{1}{0^+}=+\infty$ already — applying L'H anyway ($\to\frac{-\sin x}{1}\to 0$) manufactures a wrong answer.

indeterminate forms — rearrange, then L'Hôpital

Seven indeterminate forms: $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0\cdot\infty$, $\infty-\infty$, 1^∞, 0^0, ∞^0. Only the first two take L'H directly — the rest get rearranged into a quotient first. (Not indeterminate: $\frac{\text{nonzero}}{0}$ — that just blows up.)

***Worked:* $0\cdot\infty$ — $\lim_{x\to 0^+}x\ln x$**

1. REWRITE AS A QUOTIENT
$x\ln x=\frac{\ln x}{1/x}$ — now $\frac{-\infty}{\infty}$ ✓

↓

2. L'H
$$\lim\frac{1/x}{-1/x^2}=\lim\tfrac{1}{x}\cdot(-x^2)=\lim(-x)$$ *(dividing by a fraction = multiplying by its flip)*

↓

3. SUBSTITUTE
0 *(the x wins; $\ln x$ loses — powers beat logs)*

***Worked:* 1^∞ — $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x$**

1. TAKE ln
$\ln L=\lim x\ln\left(1+\frac{1}{x}\right)$ — a $0\cdot\infty$

↓

2. QUOTIENT + L'H
$\lim\frac{\ln(1+1/x)}{1/x}$: differentiate top and bottom — top needs the chain rule:

$$\lim\frac{\frac{1}{1+1/x}\cdot\left(-\frac{1}{x^2}\right)}{-\frac{1}{x^2}}=\lim\frac{1}{1+1/x}=1$$

(the two $-\frac{1}{x^2}$'s cancel — that's why the answer is clean)

↓

3. EXPONENTIATE BACK
$L=e^1=e$ *(the fundamental limit above — now derived, not memorized)*

$\infty-\infty$: combine over a common denominator (or conjugate) first, then treat the resulting $\frac{0}{0}$.

continuity

f is continuous at $x=a$ when:
1. $f(a)$ is defined
2. $\lim_{x\to a}f(x)$ exists
3. $\lim_{x\to a}f(x)=f(a)$

Worked: find k making f continuous

$$f(x)=\begin{cases}x^2+k & x\le 2\\ 3x & x>2\end{cases}$$

1. FIND THE STITCH POINT
only $x=2$ is at risk — each piece is a polynomial elsewhere

↓

2. SET LEFT = RIGHT
$\lim_{x\to 2^-}=4+k$ $\lim_{x\to 2^+}=6$ $\Rightarrow$ $4+k=6$

↓

3. SOLVE & CHECK
$k=2$: $f(2)=6=\lim_{x\to 2}f$ ✓ *(one equation satisfies all three continuity conditions)*

intermediate value theorem (IVT)

f continuous on $[a,b]$ and N between $f(a)$ and $f(b)\Rightarrow\exists c\in(a,b)$ with $f(c)=N$. Exam form: opposite signs at the endpoints ⇒ a root between them. IVT proves existence only — it never finds c (Newton's method, below, refines it).

Worked: show $x^3-x-1=0$ has a root in $(1,2)$

1. CONTINUITY
polynomial ⇒ continuous everywhere

↓

2. SIGNS AT THE ENDPOINTS
$f(1)=-1<0 \qquad f(2)=5>0$

↓

3. CONCLUDE
0 lies between -1 and 5 ⇒ a root lives in $(1,2)$ ■

discontinuities

- **removable**: $lim_{x\to a} f$ exists, but $\neq f(a)$.
- **infinite**: $lim_{x\to a^{\pm}} f=\pm\infty$.
- **jump**: $lim_{x\to a^-} f \neq lim_{x\to a^+} f$.

One of each: $\frac{x^2-1}{x-1}$ at $x=1$ — cancels to $x+1$, limit $=2$ exists ⇒ removable (a hole, fixable by defining $f(1)=2$). $\frac{1}{x^2}$ at 0 — $\to +\infty$ ⇒ infinite. $\frac{|x|}{x}$ at 0 — left $-1 \neq$ right 1 ⇒ jump.

Which kind? — test in this order

A **removable** — the two-sided limit exists but misses $f(c)$, or $f(c)$ is undefined. A hole. Fix it by redefining one point.
B **jump** — both one-sided limits exist but disagree. Typical of a piecewise split.
C **infinite** — a one-sided limit runs to $\pm\infty$. A vertical asymptote.

THE TEST
1. Compute both one-sided limits.
2. Do they disagree? That is a jump.
3. Is either infinite? That is an infinite discontinuity.
4. Do they agree but miss $f(c)$? That is removable.

derivatives

$f'(x)=lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$

At-a-point alternate form: $f'(a)=lim_{x\to a}\frac{f(x)-f(a)}{x-a}$ — the shape behind "given $lim_{x\to 2}\frac{f(x)-5}{x-2}=3$, find $f'(2)$" trick questions.

Worked: derivative from the definition — $f(x)=x^2$

1. BUILD THE QUOTIENT
$\frac{(x+h)^2-x^2}{h}$

↓

2. EXPAND & CANCEL
$\frac{x^2+2xh+h^2-x^2}{h}=\frac{2xh+h^2}{h}=\frac{h(2x+h)}{h}=2x+h$ *(the lone h must cancel — if it doesn't, the algebra slipped)*

↓

3. LET $h\to 0$
$f'(x)=2x$

Worked: root flavor — $f(x)=\sqrt{x}$

1. MULTIPLY BY THE CONJUGATE
$\frac{\sqrt{x+h}-\sqrt{x}}{h}\cdot\frac{\sqrt{x+h}+\sqrt{x}}{\sqrt{x+h}+\sqrt{x}}$

↓

2. TOP COLLAPSES
difference of squares: top $=(x+h)-x=h$: $\frac{h}{h(\sqrt{x+h}+\sqrt{x})}=\frac{1}{\sqrt{x+h}+\sqrt{x}}$

↓

3. LET $h\to 0$
$f'(x)=\frac{1}{\sqrt{x}+\sqrt{x}}=\frac{1}{2\sqrt{x}}$

Derivative Rules

- **Power**: $\frac{d}{dx}[x^n]=nx^{n-1}$
- **Constant Multiple**: $\frac{d}{dx}[c\cdot f(x)]=c\cdot f'(x)$
- **Sum/Diff**: $\frac{d}{dx}[f\pm g]=f'\pm g'$
- **Product**: $(fg)'=f'g+fg'$
- **Quotient**: $\left(\frac{f}{g}\right)'=\frac{f'g-fg'}{g^2}$
- **Chain**: $\frac{d}{dx}[f(g(x))]=f'(g(x))\cdot g'(x)$

Combined — the real exam item: $\frac{d}{dx}[x^2\sin(3x)]$ — product rule outside, chain inside: $2x\sin(3x)+x^2\cdot 3\cos(3x)=2x\sin(3x)+3x^2\cos(3x)$.

Common Derivatives

- $\frac{d}{dx}[c]=0, \quad \frac{d}{dx}[x]=1$
- $\frac{d}{dx}[x^n]=nx^{n-1}$
- $\frac{d}{dx}[e^x]=e^x, \quad \frac{d}{dx}[a^x]=a^x\ln(a)$
- $\frac{d}{dx}[\ln(x)]=\frac{1}{x}, \quad \frac{d}{dx}[\log_a(x)]=\frac{1}{x\ln(a)}$
- $\frac{d}{dx}[\sin(x)]=\cos(x)$
- $\frac{d}{dx}[\cos(x)]=-\sin(x)$
- $\frac{d}{dx}[\tan(x)]=\sec^2(x)$
- $\frac{d}{dx}[\csc(x)]=-\csc(x)\cot(x)$
- $\frac{d}{dx}[\sec(x)]=\sec(x)\tan(x)$
- $\frac{d}{dx}[\cot(x)]=-\csc^2(x)$
- $\frac{d}{dx}[\arcsin(x)]=\frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}[\arccos(x)]=\frac{-1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}[\arctan(x)]=\frac{1}{1+x^2}$

tangent line

At point $(a,f(a))$: $y-f(a)=f'(a)(x-a)$

Worked: tangent to $f(x)=x^2$ at $a=3$

1. HEIGHT
$f(3)=9$

↓

2. SLOPE
$f'(x)=2x \;\to\; f'(3)=6$

↓

3. POINT–SLOPE
$y=9+6(x-3)=9+6x-18=6x-9$

implicit differentiation

When y is tangled with x, don't solve for y — differentiate both sides in x, treating y as a function of x: every y-term picks up a $\frac{dy}{dx}$ by the chain rule. Then collect and solve.

Worked: folium $x^3+y^3=6xy$ — slope at $(3,3)$

1. DIFFERENTIATE BOTH SIDES
$3x^2+3y^2y'=6y+6xy'$ *(right side: product rule on $6xy$)*

↓

2. COLLECT y' TERMS
$y'(3y^2-6x)=6y-3x^2$

↓

3. SOLVE
$y'=\frac{6y-3x^2}{3y^2-6x}=\frac{2y-x^2}{y^2-2x}$

↓

4. PLUG THE POINT LAST
at $(3,3)$: $\frac{6-9}{9-6}=-1$

The multivariable shortcut $\frac{dy}{dx}=-F_x/F_y$ (Vector Calculus) compresses exactly this grind — two partials, one ratio, same answer.

logarithmic differentiation

Use when the variable sits in the exponent (x^x) or the function is a big product/quotient of powers. Take ln of both sides, expand with log rules, differentiate implicitly, multiply back by y.

Worked: $y=x^x$

1. TAKE ln
$\ln y = x\ln x$

↓

2. DIFFERENTIATE
implicit left, product rule right: $\frac{y'}{y} = 1\cdot\ln x + x\cdot\frac{1}{x} = \ln x + 1$

↓

3. MULTIPLY BACK BY y
$y' = x^x(\ln x+1)$ *(not $x\cdot x^{x-1}$ — the power rule needs a constant exponent)*

Product-of-powers flavor: $y=\frac{(x^2+1)^3(x-4)^5}{\sqrt{x}}$:
$\ln y = 3\ln(x^2+1)+5\ln(x-4)-\frac{1}{2}\ln x$; differentiate term by term (chain rule inside each ln): $\frac{y'}{y} = 3\cdot\frac{2x}{x^2+1}+5\cdot\frac{1}{x-4}-\frac{1}{2}\cdot\frac{1}{x} \Rightarrow y' = y\left[\frac{6x}{x^2+1}+\frac{5}{x-4}-\frac{1}{2x}\right]$ — no nested product/quotient rules needed.

derivative of an inverse function

$(f^{-1})'(b) = \frac{1}{f'(a)}$ where $f(a)=b$: mirrored points have reciprocal slopes — you never need a formula for f^{-1} itself.

Worked: $f(x)=x^3+2x+1$, ***find*** $(f^{-1})'(4)$

1. FIND a WITH $f(a)=4$
try small values: $f(1)=1+2+1=4$ ✓ $\Rightarrow$ $a=1$

↓

2. SLOPE OF f THERE
$f'(x)=3x^2+2 \rightarrow f'(1)=5$

↓

3. RECIPROCATE
$(f^{-1})'(4)=\frac{1}{5}$

higher-order derivatives

$f''=(f')'$, then f''', $f^{(4)}$, …, $f^{(n)}$; Leibniz $\frac{d^n y}{dx^n}$. Meaning ladder: f' rate, f'' concavity / acceleration, f''' jerk.
$f=x^4-3x^2$: $f'=4x^3-6x$, $f''=12x^2-6$, $f'''=24x$, $f^{(4)}=24$, $f^{(5)}=0$ — polynomials differentiate to death.
Patterns: $\frac{d^n}{dx^n}e^{2x}=2^n e^{2x}$; $\sin x$ cycles with period 4: $\frac{d^{50}}{dx^{50}}\sin x$ — $50 \bmod 4 = 2 \Rightarrow -\sin x$.

differentiability vs continuity

Differentiable $\Rightarrow$ continuous. The converse fails. Failure shapes at a point: **corner** ($|x|$), **cusp** ($x^{2/3}$), **vertical tangent** ($x^{1/3}$) — plus any discontinuity.

Worked: $|x|$ ***at*** 0 ***— continuous, not differentiable***

1. CONTINUOUS
$\lim_{x\to 0}|x| = 0 = f(0)$ ✓

↓

2. SLOPE FROM THE LEFT
$\lim_{h\to 0^-}\frac{|h|}{h} = -1$

↓

3. SLOPE FROM THE RIGHT
$\lim_{h\to 0^+}\frac{|h|}{h} = +1$

↓

4. COMPARE
$-1 \neq 1 \Rightarrow$ no derivative: a corner — the one-sided *slopes* must agree, exactly like one-sided limits

critical points and extrema

A **critical point** is a value c in the domain of f where $f'(c)=0$ or $f'(c)$ is undefined. Don't forget the undefined branch: $|x|$ and $x^{2/3}$ both have a critical point at 0 with no horizontal tangent.
The running example for this whole block: $f(x)=x^3-3x^2+1$, $f'=3x^2-6x=3x(x-2)$, $f''=6x-6$ — critical points $x=0,2$.

First Derivative Test

At critical point c:
- f': $+ \rightarrow -$: local max at c
- f': $- \rightarrow +$: local min at c
- no sign change: neither

$f'>0 \Rightarrow f$ increasing; $f'<0 \Rightarrow$ decreasing. **Sign chart mechanic**: mark every critical point (and domain break) on a number line, test one point per interval, read the signs.

Worked: sign chart for $f'=3x(x-2)$

1. MARK THE CRITICAL POINTS
number line split at $x=0$ and $x=2$: three intervals

↓

2. TEST ONE POINT EACH
$f'(-1)=+9$, $f'(1)=-3$, $f'(3)=+9 \Rightarrow$ signs $+\,-\,+$

↓

3. READ THE CHART
inc on $(-\infty,0)$, dec on $(0,2)$, inc on $(2,\infty)$; heights from f itself: $f(0)=1$, $f(2)=8-12+1=-3$: local max $(0,1)$, local min $(2,-3)$

Second Derivative Test

If $f'(c)=0$:
- $f''(c)>0$: local min at c
- $f''(c)<0$: local max at c
- $f''(c)=0$: inconclusive

Same f, faster: $f''=6x-6$: $f''(0)=-6<0 \Rightarrow$ max at 0; $f''(2)=6>0 \Rightarrow$ min at 2 — agrees with the sign chart. Inconclusive case: $f=x^4$ at 0 has $f''(0)=0$ yet is a min — fall back to the First Derivative Test.

Closed Interval Method (abs extrema on $[a,b]$*)*

Evaluate f at all critical points in (a,b) and both endpoints. Largest = abs max. Smallest = abs min.

Worked: abs extrema of $f=x^3-3x^2+1$ *on* $[-1,4]$

1. CRITICAL POINTS INSIDE
$x=0,2$ — both in $(-1,4)$ ✓

↓

2. EVALUATE ALL FOUR CANDIDATES
$f(-1)=-3$, $f(0)=1$, $f(2)=-3$, $f(4)=17$

↓

3. COMPARE
abs max 17 at $x=4$ (an endpoint!); abs min -3 at $x=-1$ *and* $x=2$ (a tie)

Both classic traps in one example: the max lived at an endpoint (the local max $f(0)=1$ lost), and ties are allowed — report every location.

concavity and inflection

- $f''(x)>0$: concave up (holds water)
- $f''(x)<0$: concave down (sheds water)
- **inflection**: f'' changes sign

Running example: $f''=6x-6=0$ at $x=1$, sign flips $- \rightarrow + \Rightarrow$ inflection $(1,-1)$: concave down before, up after. $f''=0$ alone is not enough: $f=x^4$ has $f''(0)=0$ with no sign change — no inflection.

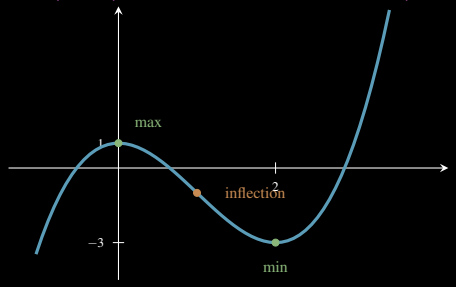
curve sketching — the synthesis

The ordered checklist (each step feeds the next):

1. **Domain** (three blockers) **2. Intercepts** **3. Symmetry** (even/odd/periodic)
4. **Asymptotes** (VA/HA/slant procedure above)
5. f' **sign chart**: inc/dec + local extrema
6. f'' **sign chart**: concavity + inflection
7. **Plot**: extrema, inflections, asymptotes, then connect respecting the two charts

Worked: sketch $f=x^3-3x^2+1$ — assembling the block's results

Domain $\mathbb{R}$; y-intercept $(0,1)$; no asymptotes (polynomial). f' chart: $+-+$ $\Rightarrow$ max $(0,1)$, min $(2,-3)$. f'' chart: $-+\Rightarrow$ inflection $(1,-1)$.



mean value theorem (MVT)

If f continuous on $[a,b]$, differentiable on (a,b): $\exists c\in(a,b)$ with $f'(c)=\frac{f(b)-f(a)}{b-a}$

Rolle's: if $f(a)=f(b)$, $\exists c$ with $f'(c)=0$.

Worked: find the MVT's c — $f=x^2$ on $[0,3]$

1. AVERAGE SLOPE
$\frac{f(3)-f(0)}{3-0}=\frac{9-0}{3}=3$

↓

2. SET $f'(c)$ EQUAL
$2c=3$

↓

3. SOLVE & VERIFY MEMBERSHIP
$c=\frac{3}{2}$ — and $\frac{3}{2}\in(0,3)$ ✓

Bound flavor (the other exam use): $f(0)=1$ and $f'(x)\le 2$ everywhere. MVT on $[0,4]$: $\frac{f(4)-f(0)}{4-0}=f'(c)\le 2\Rightarrow f(4)-f(0)\le 8\Rightarrow f(4)\le 1+8=9$ — MVT turns a slope cap into a value cap.

extreme value theorem (EVT)

f continuous on $[a,b]\Rightarrow f$ attains abs max and abs min on $[a,b]$.
Both hypotheses are load-bearing: $f=x$ on the *open* $(0,1)$ has no max (never reaches 1); $f(x)=1/x^2$ for $x\neq 0$ with $f(0)=0$, on $[-1,1]$ — defined everywhere but discontinuous at 0 — has no max. This is how EVT is tested — spot the broken hypothesis.

linear approximation & differentials

The tangent line used as a calculator: near a, $f(x)\approx L(x)=f(a)+f'(a)(x-a)$. Differentials: $dy=f'(x)\,dx$ estimates the change in f from a wiggle dx. Pick $a=$ the nearest nice value.

Worked: estimate $\sqrt{4.1}$

1. NAME f AND THE NICE BASE
$f=\sqrt{x}$, $a=4$: $f(4)=2$

↓

2. SLOPE AT THE BASE
$f'=\frac{1}{2\sqrt{x}}\to f'(4)=\frac{1}{4}$

↓

3. RIDE THE TANGENT
$L(4.1)=2+\frac{1}{4}(0.1)=2.025$ *(true 2.02485; over-estimate because $f''<0$ — tangent sits above a concave-down curve)*

Same move, two variables: the $\sqrt{3.06^2+3.92^2}$ example in Vector Calculus.

related rates

Quantities linked by an equation, all moving in time. Differentiate the relation with respect to t first, substitute values last — a substituted number has derivative 0 and kills its term.

THE PROCESS
1. Draw the picture and name every changing quantity.
2. Write the equation that ties them together. Geometry, not calculus.
3. Differentiate both sides with respect to t. Every variable gets a $\frac{d}{dt}$ by the chain rule.
4. Substitute the instant — the given values and rates. Substitute only now, never in step 2.
5. Solve for the unknown rate.

Step 4 is where marks are lost. A quantity that changes must stay a variable through the differentiation. If you plug in its number first, its derivative disappears.

Which equation ties them together?

the picture shows	the equation
a right triangle, ladder shadow	$a^2+b^2=c^2$
similar triangles	a ratio of matching sides
a cone or a cylinder filling	V formula, then eliminate a variable with a ratio
an angle changing	$\tan\theta=\frac{\text{opp}}{\text{adj}}$
area or volume growing	the area or volume formula

Worked: sliding ladder

10 ft ladder, base slides away at 1 ft/s. How fast does the top drop when the base is 6 ft out?

1. RELATION (draw it)
$x^2+y^2=100$ *(Pythagoras — true at every instant)*

↓

2. DIFFERENTIATE IN t
$2x\frac{dx}{dt}+2y\frac{dy}{dt}=0$

↓

3. FILL IN THE SNAPSHOT
$x=6\Rightarrow y=8$ (Pythagoras again); $\frac{dx}{dt}=1$

↓

4. SOLVE FOR THE ASKED RATE
$\frac{dy}{dt}=-\frac{x}{y}\frac{dx}{dt}=-\frac{6}{8}=-0.75$ ft/s *(negative = dropping, as it must)*

One-liner flavor: circle with $\frac{dr}{dt}=2$ at $r=5$: relation $A=\pi r^2$; differentiate: $\frac{dA}{dt}=2\pi r\frac{dr}{dt}=2\pi(5)(2)=20\pi$.

optimization (one variable)

Constraint kills a variable, then it's a Closed-Interval / derivative-test problem: write the objective, substitute the constraint, state the domain, find critical points, confirm max/min, answer the question asked.

THE PROCESS
1. Name the quantity to maximize or minimize. Write it as an equation.
2. Find the constraint — the second relationship in the problem.
3. Substitute the constraint to reduce to one variable.
4. State the domain. A length cannot be negative.
5. Differentiate, set to zero, and solve.
6. Test the candidates and the endpoints, then answer the question asked.

Read which quantity is given and which is free. "Fence 400 ft, greatest area" makes the perimeter the constraint and the area the objective. "Area 400, least fence" swaps them. The recipe does not change — only the roles.

Step 6 decides the answer's form: report the dimensions if asked for dimensions, and the value if asked for the maximum.

Worked: river fence — 400 ft, three sides, max area

1. OBJECTIVE + CONSTRAINT
$A=xy$ with $2x+y=400$ *(x = the two sides, y = the one parallel to the river)*

↓

2. ELIMINATE
$y=400-2x$: $A(x)=x(400-2x)=400x-2x^2$, domain $0<x<200$

↓

3. CRITICAL POINT
$A'=400-4x=0 \Rightarrow x=100$

↓

4. CONFIRM & ANSWER
$A''=-4<0$ ✓max: $x=100$, $y=200$, $A_{max}=20{,}000$ ft^2

Same recipe, two variables: the open-top box pair in Vector Calculus (min material / max volume).

rectilinear motion (1-D)

$s(t) \xrightarrow{d/dt} v(t) \xrightarrow{d/dt} a(t)$; **speed** $=|v|$; **at rest** $\Leftrightarrow v=0$; speeding up $\Leftrightarrow$ v and a share a sign; total distance $\neq$ displacement.

Worked: $s=t^3-6t^2+9t$, $t \geq 0$

1. DIFFERENTIATE TWICE
$v=3t^2-12t+9=3(t-1)(t-3)$ $\quad a=6t-12$

↓

2. AT REST / DIRECTION
$v=0$ at $t=1,3$; sign chart: right on $(0,1)$, left on $(1,3)$, right on $(3,\infty)$

↓

3. SPEEDING UP — SIGNS AGREE
$a=0$ at $t=2$: v,a both $-$ on $(1,2)$, both $+$ on $(3,\infty) \Rightarrow$ speeding up on $(1,2)\cup(3,\infty)$

↓

4. EVALUATE s AT THE KEY TIMES
$s(0)=0$, $s(1)=1-6+9=4$, $s(3)=27-54+27=0$, $s(4)=64-96+36=4$

↓

5. DISPLACEMENT vs DISTANCE on $[0,4]$
displacement $=s(4)-s(0)=4-0=4$
distance $=|4-0|+|0-4|+|4-0|=4+4+4=12$ *(split at the turnarounds $t=1,3$)*

Antiderivative flavor: $a=-32$, $v(0)=48$, $s(0)=0$: $v=-32t+48$, $s=-16t^2+48t$; max height where $v=0$: $-32t+48=0 \Rightarrow t=1.5$; $s(1.5)=-16(2.25)+48(1.5)=-36+72=36$ ft.

Newton's method

Follow the tangent line to its x-intercept, repeat: $x_{n+1}=x_n-\dfrac{f(x_n)}{f'(x_n)}$. IVT locates the root; Newton refines it. Fails if $f'(x_n)=0$ or the seed is far off.

Worked: $\sqrt{2}$ ***via*** $f(x)=x^2-2$, $x_0=2$

1. THE ITERATION
$x_{n+1}=x_n-\dfrac{{x_n}^2-2}{2x_n}$

↓

2. RUN IT
$x_1=2-\frac{4-2}{4}=1.5$; $x_2=1.5-\frac{1.5^2-2}{2(1.5)}=1.5-\frac{0.25}{3}=1.41\overline{6}$; $x_3=1.41\overline{6}-\frac{0.00694}{2.83}$ $\approx$ 1.414216

↓

3. COMPARE
$\sqrt{2}=1.414214$ — 6 digits in 3 steps (roughly doubles correct digits per step)

Calc I common pitfalls

- $0/0$ is a starting gun, not an answer — algebra first, L'H second, and re-check the form before each reapplication.
- L'H differentiates top and bottom separately — never the quotient rule.
- Related rates: differentiate first, substitute last — a plugged-in number differentiates to 0.
- Implicit: every y-term needs $\frac{dy}{dx}$ — forgetting one is the classic slip.
- Closed interval method: endpoints are candidates — the interior test points don't see them.
- $\sqrt{x^2}=|x|$ — at $-\infty$ that's $-x$; sign flips.
- Speeding up needs v AND a — same sign, not just $a>0$.
- $f''(c)=0$ decides nothing — fall back to the first-derivative sign chart.

Legend

Color & notes
functions $f,g,\sin$
variables x,y,n
numbers 0–9
operators $+-=$
delimiters $()\,[]$
titles in white
labels · results · pitfalls
$\boxed{u}$ / $\boxed{du}$ substitution

Greek
constants (fixed values):
π ≈ 3.14159
γ fluid wt density
ρ weight per length
variable & operators:
θ angle (varies)
Δ "change in"
Σ sum of terms
notation, not values:
$\Theta\,\Omega\,\omega$ growth rate, like O

Calculus & relations
$\int$ integral
$\sum$ sum
lim limit
∞ infinity
$\to$ approaches
$\Rightarrow$ implies
$\Leftrightarrow$ iff
$\sqrt{\ }$ root
$\approx$ approx
$\neq < \leq > \geq$ compare
$\pm \mp$ plus / minus
$\cdot \times \div$ mult / div
$\circ$ compose

Vectors
$\langle\rangle$ components
$\|\vec{v}\|$ magnitude
$\vec{v}$, $\hat{v}$ vector, unit
$\mathbf{ijk}$ basis
$\vec{u}\cdot\vec{v}$ dot product
proj projection

Logic & sets
$\forall$ for all
$\exists$ exists
$\neg$ not
$\wedge \vee$ and / or
$\oplus$ xor / sym. diff
$\in \notin$ in / not in
$\subseteq \subset$ subset
$\cup \cap$ union / inter.
$\setminus$ set minus
$\equiv$ equiv / congruent
$\mid$ divides / $|x|$ / s.t.
$\lceil\ \rceil$ ceiling
$\binom{n}{k}$ choose
deg degree
mod modulo